Mst Afsana Mimi

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Research Interests

Agentic AI, Bayesian Statistics, Probabilistic Modeling, Machine Learning, Statistical Inference, Computational Statistics.

Current Project

Predictive Query over Relational Databases. Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data. The long-term objective is to build a benchmark for evaluating agentic systems on realistic database tasks.

Education

University of Massachusetts Amherst, Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

- **Advisor:** Lulu Kang
- **Project: Gaussian Process Regression with Fairness Constraint** Developing fairness-aware Gaussian Process regression models for datasets involving both qualitative and quantitative variables. The project focuses on incorporating fairness constraints into probabilistic prediction models to reduce bias across protected groups while preserving predictive accuracy. Relevant methods include probabilistic modeling, kernel-based learning, constrained optimization, and computational implementation.

University of Dhaka, Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

- **Advisor:** Md Hasinur Rahaman Khan
- **Project: Variable Selection for Censored Data Using Modified Correlation Adjusted Correlation (MCAR) Scores.** Developed an MCAR feature-selection method for censored high-dimensional data using AFT models. Evaluated predictive performance through simulations and DLBCL microarray data.

University of Dhaka, Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Experience

Research Assistant, Data Systems Group, DREAM Lab, Manning College of Information and Computer Sciences, University of Massachusetts Amherst – Amherst, MA, USA. Sep 21, 2026 – Present
Project: *Relational Predictive Query Processing*.
Supervisor: Marco Serafini.

Adjunct Faculty, Department of Psychological and Brain Sciences, University of Massachusetts Amherst – Amherst, MA, USA. Sep 2026 – Present

Teaching Assistant, Department of Mathematics and Statistics, University of Massachusetts – Massachusetts, USA. Sep 2023 – May 2026

Research Assistant, Department of Computer Science and Engineering, University of Dhaka – Dhaka, Bangladesh. July 2021 – June 2022
Project Name: *Development of A Robust Bangla Language Model and a Bangla NLP Server (Funded by Centennial Research Grant, University of Dhaka)*.

Research Assistant, Institute of Statistical Research and Training, University of Dhaka – Dhaka, Bangladesh. July 2018 – June 2019
Project Name: *Poverty and Inequality Mapping in Bangladesh.*

Technical Skills

Programming: Python, R, MATLAB, C, SQL
Frameworks & Libraries: PyTorch, PyTorch Geometric
Databases & AI Systems: Relational databases, LLMs, agentic AI systems
Machine Learning & Data Mining: Predictive modeling, relational machine learning
Statistical Modeling: Gaussian Process Regression, probabilistic modeling, fairness-aware machine learning, Bayesian statistics, high-dimensional data analysis
Statistical Software: SPSS, Stata, SAS

Awards and Scholarship

- **Conference Award for Scientists (CASc)**, International Society for Clinical Biostatistics (ISCB), 2019.
- **National Science and Technology (NST) Fellowship Award** by Education Board of Bangladesh for MS thesis.
- **Dhaka University Merit Scholarship** for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra